Đinh Trung Quốc Anh

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RESEARCH INTERESTS

Star Formation History of Massive Galaxies, Dusty Star-Forming Galaxies and Galaxy Morphology at High Redshifts.

EDUCATION

International University, Vietnam National University - Ho Chi Minh City Vietnam
B.Eng in Space Engineering Sep, 2022 - Oct, 2026
All academic activities are in English
Cumulative GPA: 92.5/100 (3.87/4.00)
Thesis Title: Star Formation History of Massive Galaxies at z $>$ 3, Using Non-parametric & Parametric Models
Advisor: Dr. Abdurro'uf (Indiana University)

RESEARCH EXPERIENCE

Independent Research Project: Spatially Resolved Star Formation History of Dusty Merging Galaxies at z $>$ 2 from JWST+ALMA
Advisor: Dr. Abdurro'uf (Indiana University) Aug, 2026 - Present

- **Goal**: Understanding the impact of "wet" mergers on the star formation history (SFH) at Cosmic Noon.
- Adopted the close-pair detection and visual inspection to make a sample of 6 merging galaxies at z $>$ 2.
- Conducted AGN subtraction by comparing the performances of galaxy-only model fitting and AGN model fitting on the central point-source's SED.
- Used piXedfit for pixel binning & Bagpipes/Dense Basis for fitting spatially resolved SED.
- Analyzed JWST/NIRSpec + the dust continuum from ALMA to investigate the interplay of dust & gas in these extreme systems.

2026 Summer Internship: Morphological Properties of Galaxies in a z = 4.53 Extended Galaxy Structure
Advisors: Prof. Arianna S. Long & PhD Student Nguyễn Bình (University of Washington) Jul, 2026 - Present

- **Goal**: Quantifying the morphology of over 200 galaxies in an extended galaxy structure at z = 4.53.
- Used statmorph to derive non-parametric and parametric morphological parameters in F277W + F444W filters.
- Analyzed the morphological relations in the extended galaxy structure such as the Gini-M20, the size-mass, the axis ratio - Sérsic index - effective radius relations.
- Classified the galaxies into early-, late-, merger- types using both Gini-M20 and CAS systems.

Bachelor Thesis: Star Formation History of Massive Galaxies at z $>$ 3, Using Non-parametric & Parametric Models
Advisor: Dr. Abdurro'uf (Indiana University) Jan, 2026 - Jun, 2026

- **Goal**: Constraining the star formation history (SFH) of a sample of 80 massive galaxies at z $>$ 3 and comparing the effects of non-parametric/parametric SFH models.
- Used Bagpipes for SED fitting, with 3 SFH models (Delayed Exponential, Double-Power Law, Non-parametric using Continuity priors).
- Compared the effects of 3 models on the SFHs and the look-back times at 25%, 50% and 75% of mass formed.
- Detected a subsample of 18 massive quiescent galaxies and 4 rejuvenating candidates based on their non-parametric SFHs and derived their formation, quenching, rejuvenating timescales.

Final-year Research Project: Measuring Brightness & Fitting the Spectral Energy Distribution of a Lensed Galaxy at z = 3.29
Advisor: Dr. Phạm Tuấn Anh (Vietnam National Space Center) Sep, 2025 - Jan, 2026

- **Goal**: Measuring the flux and SED-fit properties of a gravitationally lensed, high-redshift, Lyman-alpha emitting galaxy P198.

- Used SAOImageDS9 and Photutils to measure photometry of HST and JWST images.
- Conducted SED fitting using CIGALE and examined 2 SFH models: sfhdelayed and sfh2exp.

Third-year Research Project: Exploring the Interactions between Martian Crustal Magnetic Field and Solar Winds with SWEA/MAVEN
Advisor: Dr. Jaeheung Park (Korea Astronomy and Space Science Institute) Dec, 2024 - Jun, 2025

- **Goal**: Visualizing the altitude-resolved martian magnetic field's 2D topology and its correlation with the inward/outward flows of solar-wind electrons/photoelectrons.
- Calculated the shape parameter for topology classification with SWEA/MAVEN data.
- Identified the direction of electron populations with pitch angle & magnetic field's radial component.

2024 Summer Internship: Statistical Analysis of Ionospheric Irregularity Distribution Using Swarm Data
Advisors: Dr. Wookyoung Lee (Korea Astronomy and Space Science Institute) & Dr. Hyosub Kil (Johns Hopkins University) Jul, 2024 - Sep, 2024

- **Goal**: Identifying the scintillations at the high-latitude regions during summer of each hemisphere.
- Calculated and defined threshold for S parameter - an indicator for ionospheric irregularity.
- Plotted daytime/nighttime distributions of S parameter and plasma density in a year using Swarm LP data.

2023 Summer Internship: Front-end Testing and Deployment of BURSTT
Advisor: Dr. Chao-Te Li (The Academia Sinica Institute of Astronomy and Astrophysics) Jul, 2023 - Sep, 2023

- **Goal**: Understanding the front-end design for BURSTT radio telescope to detect Fast Radio Bursts.
- Assembled the Low-Noise Amplifier & Deployed an 2x16 antenna array at Fushan site with BURSTT team.
- Modelled a simple antenna model with Ansys-HFSS.

OBSERVING EXPERIENCE

Quantifying the Rotation of Saturn with Doppler Spectroscopy
Advisor: Dr. Phan Thanh Hiền (University of Science & Technology of Hanoi) Oct, 2026

- Observed the tilted absorption lines of Saturn's spectrum with a 60-cm telescope at Quy Nhon Observatory.
- Reduced the spectrocopic data with Python and Demetra software; Measured the rotational velocity of Saturn by the Doppler shift of the H-alpha line.

Multicolor Observations of Globular Cluster NGC 2419 Using The 80-cm Twin Telescope (Teide Observatory, Spain)
Advisor: Prof. Phan Bảo Ngọc (International University, VNU-HCMC) Dec, 2024

- Submitted a proposal for remote observation of NGC 2419 in the SDSS bands u', g', r', i', z'.
- Plotted the color-color, color-magnitude diagrams for NGC 2419.

PUBLICATIONS

1. *A city at the edge of time: Spectroscopic confirmation of a $z = 4.53$ extended galaxy structure.*
N. Bình, A.S. Long, R. Oh, **D.T.Q. Anh,** et al. 2026, in-prep
2. *Spatially resolved star formation history of dusty merging galaxies at $z > 2$ from JWST+ALMA.*
D.T.Q. Anh, Abdurro'uf. 2026, in-prep

HONORS & AWARDS

Gold Medal for GPA achievement, International University 2026
Full Scholarship, International University 2022 - 2026
Development Fund's Scholarship, Vietnam National University - Ho Chi Minh City 2022 - 2026
Dean's List, International University 2023, 2025
Silver Honor, International Astronomy and Astrophysics Competition 2025
Valedictorian of Entrance Exam, International University 2022
Certificate of Merit for College/University Valedictorians, Ho Chi Minh City People's Committee 2022

SKILLS

- **Programming:** PYTHON, C, MATLAB, SWIFT, SQL, HTML, CSS
- **SED-Fitting:** Integrated & Spatially Resolved SED fitting using Bagpipes, CIGALE, piXedfit, Dense-basis
- **Photometry Tools:** APT, Siril, SAO ImageDS9, Photutils, SEP
- **Galaxy Morphology Codes:** statmorph
- **Machine Learning Algorithms:** Random Forest, K-Means, SVM, DBScan, CNN
- **Antenna/Radio Simulations:** Ansys-HFSS, GNU-Radio

POSTERS & PRESENTATIONS

Non-parametric/Parametric Star Formation Histories of Massive Galaxies at z $>$ 3 Aug, 2026
University of Washington & University of Manchester Joint Group Meeting Online
See the presentation here

Exploring the Interactions between Martian Crustal Magnetic Field and Solar Winds Jan, 2026
International University's 6th Scientific Research Student Conference Ho Chi Minh City, Vietnam
See the poster here

Statistical Analysis of Ionospheric Irregularity Distribution in High-Latitude Regions Dec, 2024
International University's 5th Scientific Research Student Conference Ho Chi Minh City, Vietnam
See the poster here

PARTICIPATED ASTRONOMICAL SCHOOLS

48th International School for Young Astronomers Sep 28 - Oct 16, 2026
Organized by The International Astronomical Union (IAU) & ICISE Gia Lai, Vietnam

SOKENDAI NAOJ Asian Winter School Feb, 2025 & Feb, 2026
Organized by The Graduate Institute for Advanced Studies - SOKENDAI Online

3rd TNU/SAGI Summer School for Observational Astrophysics Jul 27 - Aug 10, 2025
Organized by Simons Astrophysics Group at ICISE (SAGI) & Tây Nguyên University (TNU) Đăk Lăk, Vietnam

SERVICES

Member of Vietnam Amateur Astronomy Network (VAAN) Aug, 2025 - Present
Class Monitor of K2022, Space Engineering 2022 - 2026
Member of Space Engineering's Admissions Counseling Team 2023 - 2026
Content Creator for 'Preparation for Entrance Exam' Program, Thanh Niên's Newspaper 2023, 2024, 2026
Head of Space Engineering's Astronomy Club (SEAC) 2023 - 2025
Co-organizer of Mid-Autumn Moon Gazing Event for World Space Week 2025 Oct, 2025
Calculus Tutor for Space Engineering Students Oct, 2025 - Dec, 2025

REFERENCES

Dr. Abdurro'uf (Department of Astronomy, Indiana University)
Email: fnuabdur@iu.edu

Prof. Arianna S. Long (Department of Astronomy, University of Washington)
Email: aslong@uw.edu

Dr. Phạm Tuấn Anh (Department of Astrophysics, Vietnam National Space Center)
Email: ptanh@vnsc.org.vn